

AI Prompt Engineering

Peace God! The info that I wanted to send you are on actual worksheets, not on my computer. So, I typed up this document on some basic AI prompting that will show you how to properly structure a prompt in order to get the output that you want. The science is that if the prompt is weak, the output will be weak. The more structured your prompt is, the better the results. People have to remember that AI is not autonomous – it still has to be directed by humans, this is not a Terminator or Matrix movie. The term for this is “Human-In-The-Loop.” Big corporations ignore this, and that’s why their enterprise systems end up fucked off! This structure will also keep you from c’ussing out the AI, which, as you’ve seen, produces nothing conducive to what you need. Most AI platforms flag curse words and do not respond properly.

The Basic Prompt Structure

Whenever you prompt AI, the following structure should be **Role → Context → Task → Format → Tone → Constraints**.

Role: Who should the AI be? When you start, think about what you’re trying to accomplish overall. So, who does the AI need to be to accomplish your goal? Always give the AI a Role, or as they say, give it a voice.

Example: You are a software developer with over 20 years of experience in object oriented Programming, with expertise in Java Programming and the Spring Boot framework....
(this is how I start prompting AI when I use it for coding assistance)

Context: What’s the situation and audience? The situation is explaining why you’re accessing the AI, and the audience is if there is a particular individual or group of people you want to target. This comes in handy when developing something in a business or office type setting.

Example: (Situation)....the ticketing system for my department is getting backed up and we need to develop a better workflow. **(Audience)** I need to improve the workflow for the Machine Repair Maintenance department.

Task: What exactly should it do? Once you give it the context, you need to tell the AI what you want it to do in order to accomplish the goal.

Example: ... Draft an apology to Marcus who received the damaged headphones, offering him to send back the damaged item and in return, we’ll send him a full refund and another pair of headphones.
(this was from a prompt assignment where the situation was a disgruntled customer who received a broken pair of headphones)

Format: How should the output be structured? This is giving the AI instructions on how you want the output to look, or how it should be presented. Depending on what you’re trying to accomplish, you might want paragraphs, a table structure, bullet-point list or a pie chart.

Example: Craft the response in an email.....
(this was the format given for the task example above)

Tone: What is the mood that the person/group should be addressed in? We know that the way you structure written words can relay a certain mood. Should it be assertive, calm, angry, etc.

Example: ... make sure to be genuine and sincere. (the tone for the above example)

Constraints: What limits or boundaries should be placed on the output? By constraining the output, you keep it from hallucinating or giving you a long, drawn-out response.

Example: Keep the email brief but don't be short with the customer. (Constraints involve how long or short you want the output to be which can be specific or a range. For example, when it comes to time, if you wanted something to be 5 minutes in length, you would need to prompt it by saying "five minutes or less" or "more than five minutes")

Prompt Chaining

Prompt Chaining is a technique you execute when using AI to conduct research. It involves creating the first prompt, then you take the output from the first prompt and use it to create a second prompt. Then, you take the output from the second prompt to create the third prompt. This is used when you want detailed information that all relates to the topic being researched.

Hallucinating

Again, AI is not autonomous, it needs human direction. But based on how the AI is trained (the LLM – Large Language Model), it can stray off course and give you output that does not relate to what you prompted it for, or it can start repeating tasks and output that it already gave you. When this happens, you have to prompt it again and remind it of what was already done, especially if the chat with the AI is long. When we encounter this in programming, we have to use a technique called Few-Shot Prompting, which is giving it strict instructions on what and what not to do, which overrides part of the Large Language Model it was trained on.

Another reason for hallucinating is that AI, by default, is culturally biased. This is because the majority of the AI platforms were created by Caucasians. The first time I prompted ChatGPT to create an image of me without uploading an image of Self, it made me a white man. When I was in the prompt engineering class, the instructor, Tailla, did the same thing and it made her a white woman. She did that on purpose to teach us about AI being culturally biased. I'm telling you this to reinforce the fact that you need to be precise when prompting. Peace!